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Terms

1. SdV = Synthetic Data vault
2. SMOTE = Synthetic minority oversampling technique.
3. CTGAN = Condition Tabular Generative Adversarial Network

SMOTE → $\cancel{x_{new} = x + \delta(x_{neighbour} - x)}$

$$\boxed{x_{new} = x + rand(0,1) * (neighbour - x)}$$

4. SMOTE-NC = ____ – nominal & continuous

5. user prediction → o/p → {0, 1, 2, 3, 4, 5}

user label = le.inverse_transform(o/p (prediction))

SGP

FLOW 1

Load Data

↓

Encode

↓

Boderline - SMOTE

↓ —— visualization ——

Normalization

↓

Baseline NN

↓

TFL

↓

Evaluate

↓

web

FLOW 2

Load Data

↓

Encode

↓

~~[illegible]~~ SMOTE-NC

↓

Normalization

↓

Baseline NN

↓

TFL

↓

Evaluate

↓

Web

Young zerba **OR** n8n

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-> Boderline SMOTEL & SMOTE-(NC)

0 -> Heat Dissipation
1 -> No Failure
2 -> overstrain.
3 -> Power Failure
4 -> Random Failure
5 -> Tool wear

-> IMP

[y-label] column (drop when using data) OR (move to a variable).

[y-label] -> in a variable X, will be used ~~to~~ inverse_transform the user's prediction.

Deepsite

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* Raw ← 77% No Failure ^

* SMOTE BRD ← 33%

* SMOTE NC ← 16%

If MODEL does NOT work as expected ←

Generate new Data ←

Because old data is generated, with (Normalizing) before SMOTE

NOW ← Apply SMOTE on [Maintenance - CSV]

then normalize Data.

→ No Generalization on Unseen Data.

Small opportunities are often the beginning of great enterprises.

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=) New Flow

Data

↓

Label encoding (categorical only)

↓

SMOTE

↓

Split (train/test)

↓

Normalize

↓

Modelling

=) Better

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Models

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⇒ Next

DL & Transfer learning.

⇒ ① ~~smote - Boderline r~~

a) ~~ML r~~

~~random-forest-99-joblib~~

b) DL r →

~~Transformer~~

x[:, None, :]

↓ Batch

:) feature-dim

) seq-length → None = 1) For Numerical.

A leader is one who knows the way, goes the way and shows the way.

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1 Models

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* SMOTE-Boderline

ML

* random_forest_99.joblib => 99.97%

DL

* dl_baseline.h5 (88.79, 27.53)

* dl_1_transformer.keras. (88.9, 23.5)

* dl_2_transformer.keras (92.97, 19.47)

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SMOTE - NC

ML

* random_forest_98.joblib => 98.06%

DL

* dl2_baseline.keras (89.29, 28.74)

* dl2_transformer.keras (93.03, 20.98)

Transformer (Contd.)

(Query, Key, Values).

→ How much each element should focus on others.

→ Attention = softmax $\left(\frac{QK^T}{\sqrt{d_k}}\right)V$

d_k → dim. of key vector

⇒ Head → Perspective on data.